

Faed Ahmed Arnob

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I have a strong passion for research and development that will advance technology's potential to solve serious issues. My research interests include but are not limited to Computer Vision, Extended Reality, Embedded Systems, Intelligent Transportation Systems & HCI. I am also trained on Machine Learning, Deep Learning, Artificial Intelligence, Virtual Reality, Augmented Reality and Mixed Reality by IIT KGP & IITG under BDSET Program.

Education

PhD in Civil Engineering major in Transportation Engineering

CGPA: 3.76/4.00

Fall 2025 - Present

The City College of New York, CUNY, 160 Convent Avenue New York, NY 10031

Current Progress: Successfully completed first-year coursework; currently developing infrastructure-based spatial computing and artificial intelligence frameworks for urban mobility.

Supervisor: Dr. Yiqiao Li - AI & Mobility Research Lab

Master of Science in Computer Science and Engineering

CGPA: 4.00/4.00 - Incomplete

2024

United International University, United City, Madani Ave, Dhaka 1212, Bangladesh

Current Progress: Discontinued in my second-to-last semester; major track was Data Science (Thesis-Based)

Bachelor of Science in Computer Science and Engineering

CGPA: 3.73/4.00

2017-21

BRAC University, 66 Mohakhali, Dhaka, Bangladesh

Undergraduate Thesis: An Efficient Traffic Management System to Detect Lane Based Rule Violation using Real-Time Object Detection

Supervisor: Md. Motaharul Islam, PhD, Co-Supervisor: Jannatun Noor

Professional Experience

Graduate Research Assistant, AI & Mobility Research Lab

August 2025 – Present

AI, Transportation, Engineering

The City College of New York, CUNY

○ Leading research on urban intersection safety, employing a multimodal AI approach to analyze infrastructure-ADAS interactions for crash risk prediction, resulting in an accepted conference paper.

○ Specializing in the processing and analysis of LiDAR and video data; developing computer vision models for 3D object detection, tracking, and scene segmentation to extract critical safety metrics from complex urban environments.

Adjunct Lecturer / Graduate Teaching Assistant, Dept. of Civil Engineering

January 2026 – Present

Teaching, Civil Eng.

The City College of New York, CUNY

○ **Fall 2026:** Co-instructing "CE 10000 - Fundamentals of CE Prb Solving," an in-person undergraduate civil engineering course.

○ **Spring 2026:** Assisted Dr. Yiqiao Li for "CE 32700 - Transportation Systems Engineering," covering principles of traffic engineering, multimodal level of service analysis, and traffic control.

Research Engineer, Advanced Intelligent Multidisciplinary Lab

June 2023 – April 2025

Augmented Reality, Development

United International University (UIU) & ULAB

○ Worked on the "Feasibility Study of the effectiveness of AR technology in assisting children with Autism Spectrum Disorder for literacy." Currently developing an educational AR app & wearable wrist device to track the stimuli of children.

Trainee, Dept. of Computer Science & Engineering

June 2022 – February 2023

AR, VR, XR, AI, ML, DL Engineer

Indian Institute of Technology (Guwahati & Kharagpur), India

○ A 6-months training with 1 month internship program fully funded by "The Government of India and Bangladesh Hi-Tech Park Authority (BHTPA)". The training focused on Augmented Reality, Virtual Reality, Mixed Reality, Extended Reality, Artificial Intelligence, Machine Learning, and Deep Learning. BDSET

Lecturer (Contractual), Dept. of Computer Science & Engineering

January 2022 – May 2022

Java, Python, Simulator

BRAC University, 66 Mohakhali, Dhaka, Bangladesh

○ Course Taught : Programming Language I (Structured Programming)

Publications

Engagement in ASD Children During ARise Learning Sessions: A Video Frame Analysis Approach -Submitted

Authors: Arnob, F.A.; Doulah, A.B.M.S.U.; Rasheduzzaman, M.; Sarker, F.; Roy, N.; Mamun, K.A.

ARise – an Augmented Reality app for Enhancing Literacy and Learning in Children with ASD -Submitted

Authors: Arnob, F.A.; Doulah, A.B.M.S.U.; Rasheduzzaman, M.; Sarker, F.; Roy, N.; Mamun, K.A.

Application of Augmented Reality Interventions for Children with Autism Spectrum Disorder (ASD) - [View Paper](#)

Authors: Doulah, A.B.M.S.U.; Rasheduzzaman, M.; Arnob, F.A.; Sarker, F.; Roy, N.; Ullah, M.A.; Mamun, K.A.

A Novel Traffic System for Detecting Lane-Based Rule Violation - [View Paper](#)

Authors: Md. Azmol Fuad, Faed Ahmed Arnob, Abu Tahir Nizam and Md. Motaharul Islam.

An Intelligent Traffic System for Detecting Lane Based Rule Violation - [View Paper](#)

Authors: F. A. Arnob, A. Fuad, A. T. Nizam, S. Barua, A. A. Choudhury and M. Islam.

Posters

An Accurate Infrastructure-Based LiDAR Framework for Precise Vehicle Length Estimation - [View Poster](#)

Authors: Faed Ahmed Arnob, Bo Shang, Yiqiao Li

Understanding Infrastructure–Advanced Driver-Assistance System Interactions in Urban Crash Risk: A Multimodal AI Approach for Intersection Safety - [View Poster](#)

Authors: Faed Ahmed Arnob, Bo Shang, Yiqiao Li, Abolfazl Karimpour

Advancing Pregnancy Wellness: Smart Technologies and Data Analysis to Reduce Stillbirth Incidences - [View Poster](#)

Authors: Faed Ahmed Arnob, Maliha Jahan Maisha, Marzia Zaman, Sameena Chowdhury, Saleha Begum Chowdhury, Ravi Vaidyanathan, Khondaker A. Mamun

Internships and Research Projects

An Accurate Infrastructure-Based LiDAR Framework for Precise Vehicle Length Estimation

City College of New York at CUNY

Spring 2026

- Developed a Trajectory-Aligned Morphological Framework to robustly extract precise bumper-to-bumper vehicle lengths from raw roadside LiDAR point clouds.
- Introduced an AI-assisted ground truth generation method leveraging Vision-Language Models (VLMs) to automatically validate LiDAR measurements against manufacturer catalogs.
- Evaluated the framework on complex urban highway traffic, achieving high accuracy with an average Mean Absolute Error (MAE) of 0.298 meters and a MAPE of 5.2% across diverse vehicle classes.

Infrastructure-ADAS Interactions in Crash Risk: A Multimodal AI Approach for Intersection Safety

The City College of New York, CUNY

Summer 2025

- Researching the interaction between road infrastructure and Advanced Driver-Assistance Systems (ADAS) to assess urban crash risk at intersections.
- Developing and applying multimodal AI models for intersection safety analysis, fusing computer vision and traffic data.
- Author of an accepted conference submission at TRB, "Understanding Infrastructure-ADAS Interactions in Urban Crash Risk: A Multimodal AI Approach for Intersection Safety."

Feasibility Study of the Effectiveness of AR Technology in Assisting Children with ASD for Literacy

University of Liberal Arts Bangladesh

Summer 2023

- Developed an AR mobile app with flashcards/book for Bangla alphabets and numbers.
- Explored the different vital responses during the learning sessions (using FitBit tracker).
- Explored the potential of the system in effective learning for children with autism through a feasibility study.

Research Internship: Bangla Paraphrasing Tool

IIT Kharagpur

Fall 2022

- Developed proper Bangla GloVe embeddings. The model was capable of accurate next-word generation in Bengali.

Traffic Monitoring System

BRAC University

Summer 2019

- Focused on 'Traffic Lane-Rule Violations', Raspberry Pi as SoC. Used Computer Vision and YOLO to detect defaulters. A complete device was developed which can be deployed on existing traffic cameras. Achieved an overall **78% accuracy** in detection of rule violation.

Technical Strengths

- **Programming Languages:** Java, Python, C#, PyTorch(basic), MySQL, JavaScript(basic), HTML/CSS, Assembly (x86, 8051)
- **Hardware Description and Modeling Languages (HDLs):** Verilog, VHDL
- **Developer Tools/Frameworks/Libraries:** TensorFlow, Scikit-learn, Unity 3D, Verilog, VHDL, Jupyter Notebooks, Git, Postman, Keras, Pandas, NumPy, Docker, VS Code.
- **Others:** L^AT_EX, Matlab, MIPS-Assembly, Proteus, Adobe Creative Tools
- **Language:** English (Fluent & IELTS Overall Band: 7.5/C1), Bangla(Native & Fluent)

Scholastic Achievements & Awards

- Awarded a competitive **PhD Fellowship** by the **Grove School of Engineering, CCNY**, which encompasses a full tuition waiver and a handsome renewable stipend per academic year for up to five years.
- Secured position in the **Vice Chancellor's List** — BRAC University for **6 semesters** as recognition for achieving a **GPA of 3.90 - 4.00** on those particular semesters.
- Secured position in the **Dean's List** — BRAC University for **2 semesters** as recognition for achieving a **GPA of 3.70 - 3.89** on those particular semesters.
- Awarded **Vice Chancellor Certificate** — **BRAC University (Residential - 46)** for High Standards of Discipline and Devotion to duties
- **Conference Paper Presenter** in International Conference on Advances in The Emerging Computing Technologies, February 2020, Islamic University of Madinah, KSA.
- Awarded **Merit Scholarship Based on BracU Academic & previous Results** and gained **70%** tuition fee waiver for 10 semesters

Seminar & Talks

Bridging Reality: Augmented Reality in Modern Robotics

University of Liberal Arts Bangladesh

June 2024

WEB AR, Unity

- Led an enriching workshop on "Bridging Reality: Augmented Reality in Modern Robotics" at IEEE ULAB Robotics and Automation Society SBC. Explored AR integration with robotics, covering fundamental concepts and real-world applications. Demonstrated developed web AR applications, sparking an interactive Q&A session. Delivered a hands-on, enlightening experience for attendees, inspiring innovation in AR and robotics.

Seminar on Augmented Reality

August 2023

University of Liberal Arts Bangladesh

AR/VR

○ Explored the captivating realm of Augmented Reality (AR) in this engaging seminar! With a strong understanding of the field, I unraveled the wonders and potentials of AR technology. From its practical applications to its influence on various industries, this seminar of mine was an enlightening one.

Extracurriculars

Society Student Member, American Society of Civil Engineers (ASCE)

Active – 2026

○ Member Number: 000012603831. Active participation in the Metropolitan Section and the Transportation and Development Institute (TDI). - asce.org

Member, ACM SIGCHI

Summer 2022 – Present

○ ACM SIGCHI is a global community of students and professionals for advancing research, education, and practical applications in Human-Computer Interaction, fostering innovation in technology's interaction with people and its impact on their lives. - <https://sigchi.org/>

Senior Counselor & Research Mentor

Spring 2018 – Present

BRAC University Leadership Development Forum

○ Research Mentor for final year students - <https://www.facebook.com/bracu.ldf>

Creative Content Creator

December 2015 – Present

YouTube

○ Where I upload my deeds and tell my stories in the creative way possible – youtube.com/@FaedAhmedArnob.

Community Ride Moderator and Community Leader

Summer 2014 – Present

BDCyclist

○ Largest Cycling community in Bangladesh - <http://bdcyclists.com/>

References

Yiqiao Li, Ph.D., Assistant Professor Dept. of Civil Engineering

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The City College of New York (CUNY)

Computer Science Program, Graduate Center

AI & Mobility Research Lab, MR 728

275 Convent Avenue, ST 109

New York, NY 10031

Abul Barkat Mollah Sayeed Ud Doulah, PhD, Associate Prof. Dept. of EEE

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University of Liberal Arts Bangladesh

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Md. Motaharul Islam, PhD, Prof. Dept. of CSE

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United International University

United City, Madani Avenue, Badda Dhaka 1212, Bangladesh.

Dr. Debasis Samanta, Associate Prof. Dept. of Computer Science & Engineering

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